

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 3 – Comparative Analysis

|                  |                                                                                                                                   |               |                                          |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------|
| Course Code:     | DSA0308                                                                                                                           | Course Title: | Natural Language Processing              |
| CO Assessed:     | CO4 – Precision                                                                                                                   | Assessment:   | Assessment Tool 3 – Comparative Analysis |
| Weightage:       | 30% of CIA                                                                                                                        |               |                                          |
| CO5 Statement:   | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) |               |                                          |
| Student Name:    | T.Megha Harsha                                                                                                                    | Reg. No.:     | 192424389                                |
| Section / Batch: | 2024                                                                                                                              | Date:         |                                          |

### Code of Conduct

*I T.Megha Harsha(192424389)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Megha

### Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

| Section / Topic                                         | Focus                                                                                                                                                                          | Question No. |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Representing Meaning & Meaning Structure of Language    | Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words. | Q1           |
| Syntax-Driven Semantic Analysis                         | Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.              | Q2           |
| Lexemes and Their Senses & Word Sense Disambiguation    | Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.          | Q3           |
| Representing Linguistically Relevant Concepts           | Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.                            | Q4           |
| Information Retrieval & Syntax-Driven Semantic Analysis | Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.                          | Q5           |

## Annexure A – Comparative Analysis

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1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

## Rubrics for Calculation

| Criteria                | Weightage (%) | Level 4 (Exemplary)                                                           | Level 3 (Proficient)                            | Level 2 (Developing)                       | Level 1 (Beginning)                      |
|-------------------------|---------------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------|
| Scope of Items          | 20%           | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant        |
| Comparison Depth        | 25%           | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison            |
| Use of Evidence         | 20%           | Supports comparison with accurate data, examples, or references               | Uses some supporting evidence with minor gaps   | Limited or weak supporting evidence        | No supporting evidence provided          |
| Critical Thinking       | 20%           | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation; lacks depth          | No meaningful interpretation             |
| Clarity of Presentation | 15%           | Well-organized, clear, and logically structured presentation                  | Generally clear with minor issues               | Somewhat unclear or poorly structured      | Disorganized and difficult to understand |

| Q. No. / Criteria Level | Scope of Items | Comparison Depth | Use of Evidence | Critical Thinking | Clarity of Presentation | Sub. Total | Total % |
|-------------------------|----------------|------------------|-----------------|-------------------|-------------------------|------------|---------|
| 1                       | 3              | 3                | 3               | 3                 | 4                       | 16         | 78      |
| 2                       | 4              | 3                | 3               | 3                 | 3                       | 16         |         |
| 3                       | 3              | 3                | 3               | 3                 | 3                       | 15         |         |
| 4                       | 3              | 3                | 3               | 3                 | 3                       | 15         |         |
| 5                       | 3              | 3                | 4               | 3                 | 3                       | 16         |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

## Q1. Constituency Parsing vs Dependency Parsing

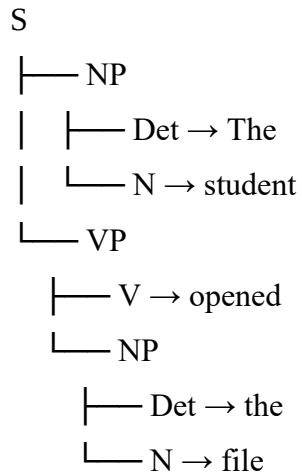
A conversational AI system needs to identify relationships between words, especially when sentences contain long-distance dependencies.

### Constituency Parsing

Constituency parsing divides a sentence into hierarchical phrases such as noun phrases (NP) and verb phrases (VP).

Example:

The student opened the file.



It shows how words form larger grammatical units.

Main focus:

Sentence



Phrases



Sub-phrases



Words

It is useful for understanding hierarchical structure.

### Dependency Parsing

Dependency parsing represents direct relationships between individual words.

For:

The student opened the file.

the dependency relationships are approximately:

opened

└─ student → subject

└─ file → object

└─ The → determiner

The verb "opened" is the central head, while other words depend on it.

Main focus:

Head Word

↓

Dependent Words

### Comparison

| Feature                      | Constituency Parsing              | Dependency Parsing                 |
|------------------------------|-----------------------------------|------------------------------------|
| Representation               | Phrase hierarchy                  | Word-to-word relationships         |
| Main units                   | NP, VP, PP                        | Head and dependent                 |
| Shows phrases                | Yes                               | Less directly                      |
| Shows direct relations       | Indirectly                        | Directly                           |
| Semantic relation extraction | Moderate                          | Very useful                        |
| Long-distance relations      | Can represent them hierarchically | Direct dependency links are useful |
| Conversational NLP           | Useful                            | Highly useful                      |
| Interpretability             | High                              | High                               |

### Example of long-distance relationship

Consider:

Which book did the student say that the teacher recommended?

The word "which book" is related to "recommended" even though several words occur between them.

Dependency parsing can directly represent the relationship:

recommended → book

This makes dependency structures useful for information extraction.

### Justification

For conversational text, dependency parsing is generally more suitable when the objective is to extract meaningful relationships such as:

Person → action

Action → object

Object → attribute

However, constituency parsing is valuable when the system needs detailed phrase structure.

### Conclusion

|                                                                 |
|-----------------------------------------------------------------|
| Dependency Parsing is more suitable for relationship extraction |
|-----------------------------------------------------------------|

while constituency parsing is better for analyzing hierarchical phrase structure. A modern conversational NLP system can benefit from both.

## Q2. Top-Down Parsing vs Earley Parsing

The system receives incomplete queries such as:

best hotels near...

and:

book a flight from Chennai to...

### Top-Down Parsing

Top-down parsing starts from the start symbol S and attempts to generate the input.

Example:

S

↓

VP

↓

V NP

↓

Book NP

The parser tries to construct a complete grammatical structure.

### Problems with incomplete input

Consider:

Book a flight from Chennai to...

The parser expects more input after "to".

A conventional top-down parser may have difficulty deciding which grammatical structure should be followed.

### Ambiguity

If several grammar rules are possible, it may try one structure, fail, backtrack, and try another.

This can cause:

More alternatives

↓

More backtracking

↓

More computation

↓

Higher response time

### Earley Parsing

Earley parsing is a dynamic-programming parsing method for CFGs.

It maintains partial parsing states and uses three major operations:

1. Prediction
2. Scanning
3. Completion

It can maintain partial structures while processing input.

For:

best hotels near...

the parser can maintain:

Query

└── hotels

└── near ...

and wait for additional input.

### Comparison

| Feature                | Top-Down                  | Earley                        |
|------------------------|---------------------------|-------------------------------|
| Starting point         | Start symbol              | Start symbol with state chart |
| Incomplete input       | Poor                      | Better                        |
| Ambiguity              | Backtracking may increase | Maintains alternatives        |
| Repeated work          | Higher                    | Reduced using chart           |
| Partial parsing        | Limited                   | Strong                        |
| Complex grammar        | Less efficient            | Suitable                      |
| Interactive NLP        | Moderate/Poor             | Better                        |
| Real-time applications | Less suitable             | More suitable                 |

### Example

Input:

Book a flight from Chennai to...

Earley parsing can maintain a partial structure:

BOOKING

└── Action → Book

└── Object → Flight

└── Source → Chennai

└── Destination → incomplete

When the user says:

Delhi

the system updates:

Destination → Delhi

rather than rebuilding everything from the beginning.

## Conclusion

Earley Parsing is more appropriate for interactive NLP

because it handles incomplete and ambiguous inputs more flexibly and supports partial parsing.

### Q3. CFG vs PCFG vs Neural Constituency Parser

A grammar-checking system needs to resolve syntactic ambiguity.

Example:

I saw the man with a telescope.

Possible interpretations:

1. I used a telescope to see the man.
2. The man had a telescope.

This is a PP attachment ambiguity.

#### 1. CFG

A Context-Free Grammar represents possible syntactic structures using production rules.

Example:

$S \rightarrow NP VP$

$VP \rightarrow V NP$

$NP \rightarrow NP PP$

$VP \rightarrow VP PP$

CFG can generate multiple valid parse trees.

#### Advantage

Highly interpretable.

#### Limitation

Basic CFG cannot naturally decide which parse is more probable.

#### 2. PCFG

A Probabilistic Context-Free Grammar adds probabilities to grammar rules.

Example:

$VP \rightarrow VP PP \quad 0.3$

$NP \rightarrow NP PP \quad 0.7$

The parser can calculate probabilities for competing structures.

Therefore:



$$P(\text{Parse} \mid \text{Sentence})$$

can be used to select the most likely interpretation.

### **Advantage**

Handles ambiguity better than basic CFG.

### **Limitation**

Performance depends on training data and grammar quality.

## **3. Neural Constituency Parser**

A neural parser uses machine-learning models to predict syntactic structures.

It can learn patterns from large datasets.

For example:

Sentence



Neural Encoder



Contextual Representation



Constituency Prediction



Parse Tree

Modern neural models can capture contextual information that simple grammar rules cannot.

### **Advantages**

- High practical accuracy
- Learns complex patterns
- Handles contextual ambiguity
- Works well with large datasets

### **Limitations**

- Less interpretable
- Requires training data
- More computationally expensive
- Difficult to explain individual decisions

## **Comparison**

| Feature | CFG | PCFG | Neural Parser |
|---------|-----|------|---------------|
|---------|-----|------|---------------|

|                      |                 |                                  |                          |
|----------------------|-----------------|----------------------------------|--------------------------|
| Grammar rules        | Yes             | Yes                              | Learned                  |
| Probability          | No              | Yes                              | Yes                      |
| Ambiguity resolution | Limited         | Good                             | Very good                |
| Interpretability     | Very high       | High                             | Lower                    |
| Training data        | Not necessarily | Usually needed for probabilities | Large datasets preferred |
| Practical accuracy   | Moderate        | Better                           | Generally highest        |
| Computational cost   | Low             | Moderate                         | Higher                   |

### Best choice

If the requirement is:

Interpretability + reasonable accuracy

then:

*PCFG*

is an excellent balance.

If maximum practical accuracy is the main objective and sufficient data/computation are available:

Neural Constituency Parser

is generally preferable.

### Conclusion

CFG provides the basic structure, PCFG adds probabilistic ambiguity resolution, and neural parsing provides strong contextual prediction.

For a grammar-checking system requiring both understandable decisions and good accuracy, a hybrid approach is ideal:

CFG

↓

PCFG

↓

Neural ranking/validation

### Q4. Feature Structures vs Traditional CFG

A multilingual NLP system must handle:

Number

Gender

Person

Tense

Case

Agreement

Traditional CFG alone cannot conveniently represent all these constraints.

### **Traditional CFG**

Example:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

The grammar allows:

The boy plays.

The boys play.

But basic CFG does not explicitly encode that:

boy  $\rightarrow$  singular

boys  $\rightarrow$  plural

plays  $\rightarrow$  singular

play  $\rightarrow$  plural

Therefore, additional mechanisms are required.

### **Feature Structures**

A feature structure attaches attributes and values to linguistic units.

Example:

NP

|— Number = Singular

|— Person = Third

└— Gender = Masculine

Verb:

VP

|— Number = Singular

|— Person = Third

└— Tense = Present

The parser can check agreement between features.

### **Example**

Correct:

The boy plays.

Features:

boy:

Number = Singular

plays:

Number = Singular

Incorrect:

The boy play.

Features:

boy:

Number = Singular

play:

Number = Plural

The feature mismatch indicates an agreement error.

### **Multilingual usefulness**

Different languages may encode:

Gender

Number

Person

Case

Tense

For example, nouns and adjectives may need to agree in gender and number.

Feature structures can represent:

Noun

└─ Gender = Feminine

└─ Number = Singular

└─ Case = Nominative

This makes them highly useful for multilingual NLP.

### **Comparison**

| Feature      | CFG     | Feature Structures |
|--------------|---------|--------------------|
| Basic syntax | Yes     | Yes                |
| Number       | Limited | Explicit           |
| Gender       | Limited | Explicit           |
| Person       | Limited | Explicit           |

|                     |           |               |
|---------------------|-----------|---------------|
| Tense               | Limited   | Explicit      |
| Agreement           | Difficult | Easy          |
| Multilingual NLP    | Limited   | Highly useful |
| Constraint checking | Weak      | Strong        |

## Conclusion

Feature structures extend CFG by attaching linguistic attributes to words and phrases.

Feature Structures are essential for enforcing grammatical constraints

especially in multilingual NLP systems.

## Q5. Subcategorization Frames

A voice assistant needs to understand that different verbs require different arguments.

Consider:

Give the book to John.

Sleep.

Put the file on the desk.

The verbs have different argument requirements.

### What is a subcategorization frame?

A subcategorization frame specifies the type and number of arguments that a verb requires.

It tells the NLP system:

What arguments does this verb need?

#### 1. Give

Sentence:

Give the book to John.

The verb "give" typically requires:

Agent

Theme/Object

Recipient

Frame:

GIVE

└─ Subject → Agent

└─ Object → Theme

└─ PP → Recipient

Semantic representation:

Give(Agent, Book, John)

## 2. Sleep

Sentence:

The child sleeps.

"Sleep" does not require an object.

Frame:

SLEEP

└— Subject → Experiencer/Agent

Incorrect:

The child sleeps the book.

The verb does not normally allow a direct object.

## 3. Put

Sentence:

Put the file on the desk.

"Put" requires:

Agent

Object

Location

Frame:

PUT

└— Subject → Agent

└— Object → File

└— PP → Location

Semantic representation:

Put(Agent, File, Desk)

The location phrase is important because:

Put the file.

is normally incomplete.

## Why subcategorization is important

### 1. Syntactic analysis

It helps determine whether a sentence is structurally valid.

For example:

She gave John.

may be incomplete because "give" normally requires a theme as well.

## 2. Semantic analysis

It identifies semantic roles.

Give the book to John.

becomes:

Agent → implied subject

Theme → book

Recipient → John

## 3. Voice assistants

When the user says:

Put the file on the desk.

the assistant can convert it into:

ACTION = PUT

OBJECT = FILE

LOCATION = DESK

## 4. Information extraction

Subcategorization helps extract structured information from text.

## 5. Ambiguity resolution

The verb's argument requirements provide contextual information that helps select the correct parse.

### Comparison of the three example verbs

| Verb  | Required structure           | Example                  |
|-------|------------------------------|--------------------------|
| Give  | Subject + Object + Recipient | Give the book to John    |
| Sleep | Subject                      | The child sleeps         |
| Put   | Subject + Object + Location  | Put the file on the desk |

## Conclusion

Subcategorization frames connect syntax with semantics by specifying the arguments required by a verb.

|                                                                                |
|--------------------------------------------------------------------------------|
| Subcategorization helps determine valid sentence structures and semantic roles |
|--------------------------------------------------------------------------------|

It is therefore very important for voice assistants, semantic parsing, information extraction, and conversational NLP.